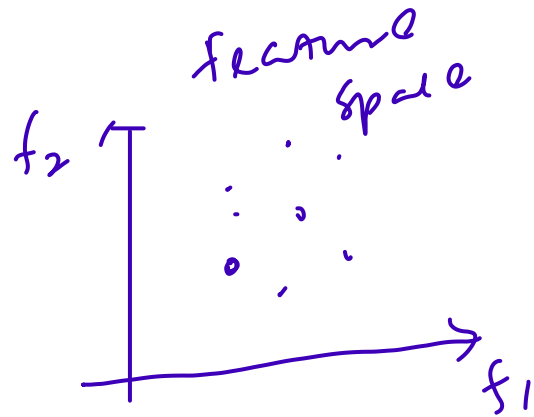


Data $\rightarrow$ Data Preprocessing $\rightarrow$ Data Mining $\rightarrow$ Knowledge

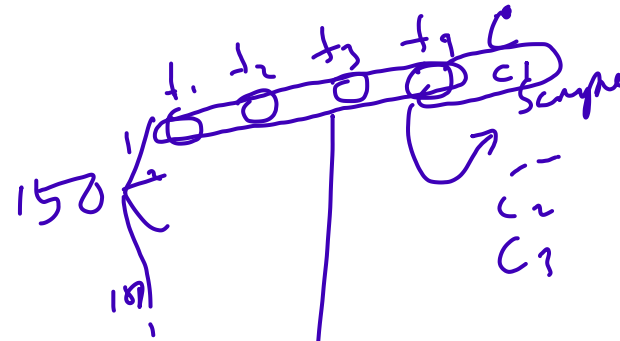
Data

Attribute/feature

Sample/Object/instance/entry/case/record/print



(f_1, f_2)



$f = [f_1, f_2, f_3, f_4]$

$\downarrow$
feature vector
dimension is 4

Types of attributes:

Nominal:

Ordinal: 1-5.

Interval: value that represents interval

Ratio: free num.
↳ discrete & continuous

Operations:

- (1) Distinctness = $\neq$
- (2) Order < >
- (3) Addition + -
- (4) Multiplication * /

Person id	Retired	Marital Status	Taxable Income	Charl Lickel
1	yes	single	125k	No
2	No	Married	70k	No
3	No	Divorced	60k	Yes

Discrete attribute: Integer

Special case: 0 & 1

Binary
attribute

* Continuous Attribute :

KNN

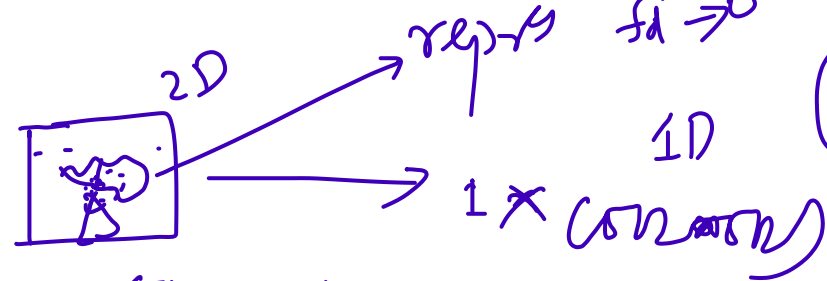
(0-1)

f_1	f_2	f_3	f_4
0	0	0	0
1	1	1	1

$\text{enc}(k) \rightarrow (0,1)$
 $(Wx + b)$
 $n \times d \quad d \times 1 \quad n \times 1$



MNIST



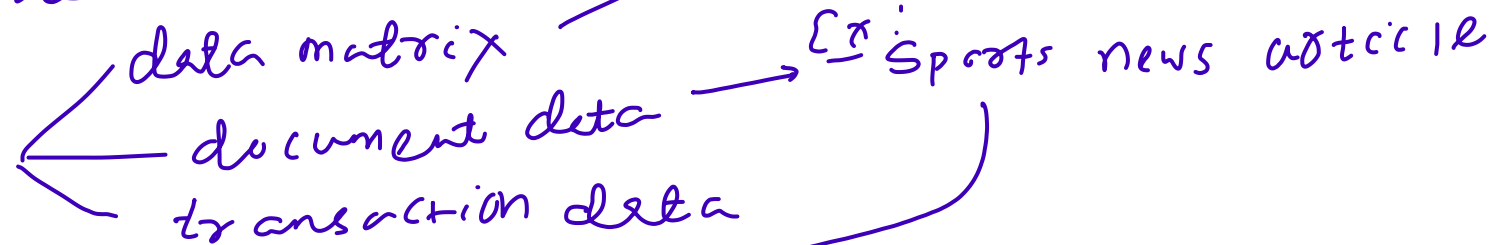
512x512

$2^9 \times 2^9 = 2^{18}$

1x784
 Digit

Types of the dataset:

① Record



$N \times d$ → no. of features
→ no. of samples

② Graph

③ Ordered

Team	Goals	Play	Ball	Score	Game	Win	Loss	Timeout	Sec.
	0	5	2	0	6	0	2	1	1

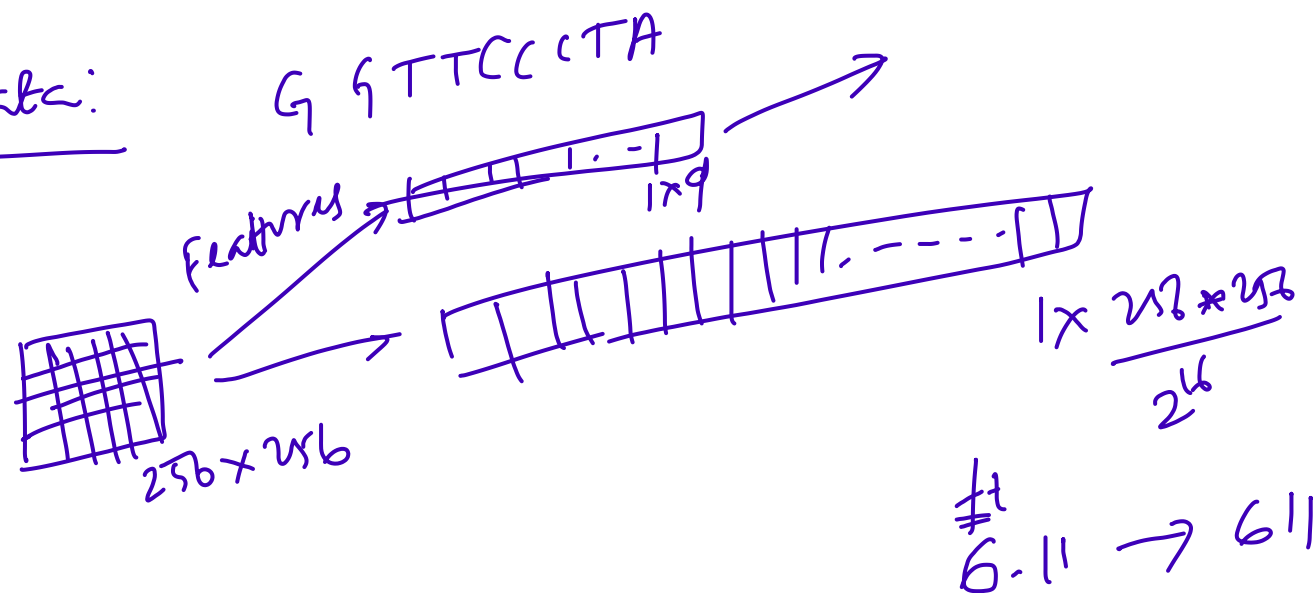
Doc 1	[3
Doc 2	[
...	:

Coke	Milk	Biscuits	Dough
1	1	0	0

TransID	Items
1	Bread, Coke, Milk
2	Butter, Biscuits, Dough
3	Coke, Biscuits, Milk

Bread
→ [1]

Ordered Data:



Data Quality:

- Problems:
- Noise & Outliers
 - Missing values
 - Duplicate data

Bias Variance Tradeoff:

①
Train error: 1%
Validation error: 12%
(Overfitting)

low bias & high variance

② 15%
17%
(underfitting)

High bias
&
low variance

$$\text{Bias}^2 = E\{[\hat{y} - y]^2\}$$

$$\text{var} = E\{[\hat{y} - \bar{\hat{y}}]^2\}$$

③

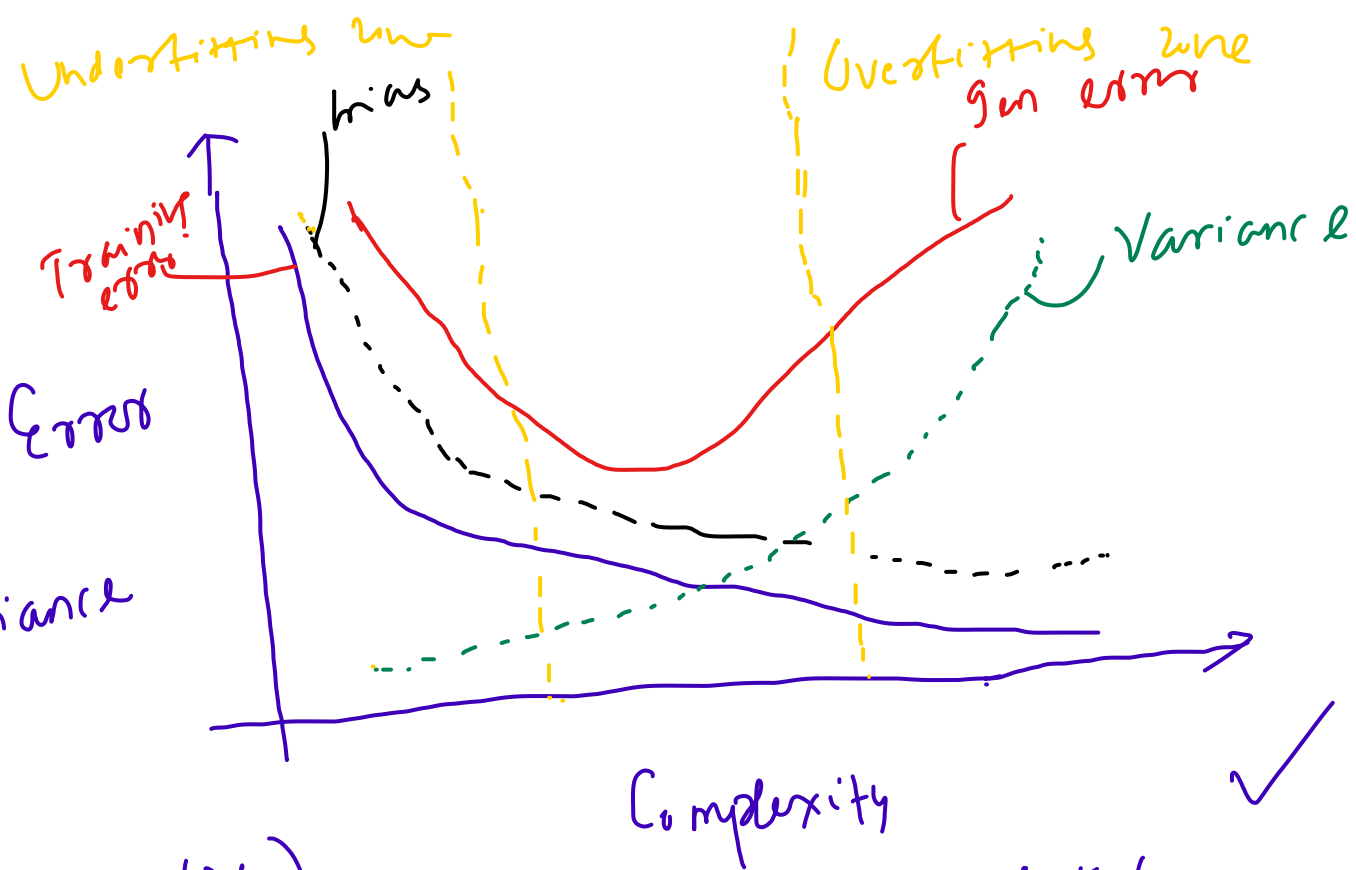
(X) 15%
30%

High bias
&
high variance

④

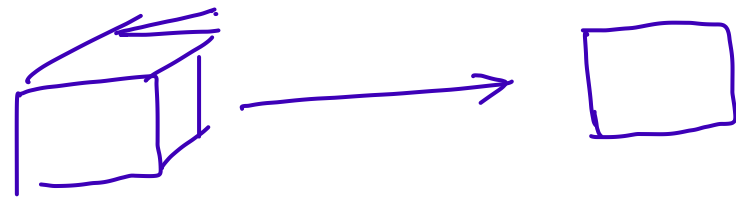
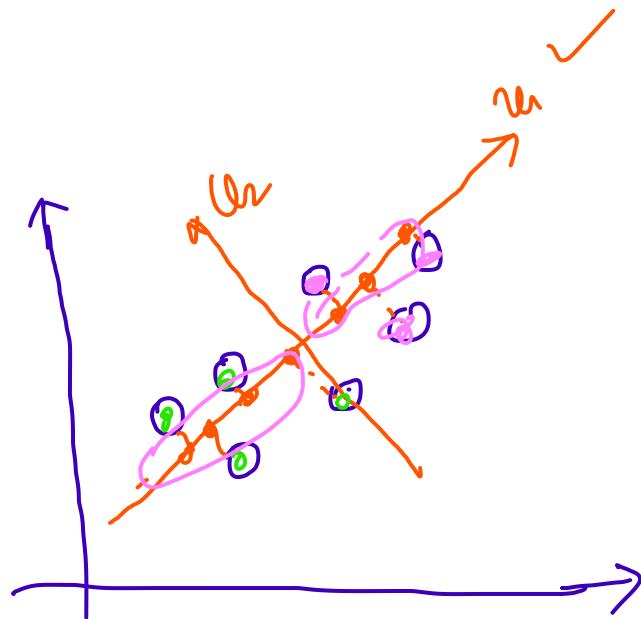
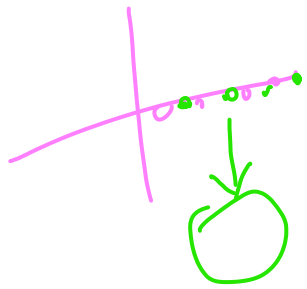
0.5% (proper fit)

1%
low bias
&
low variance ✓

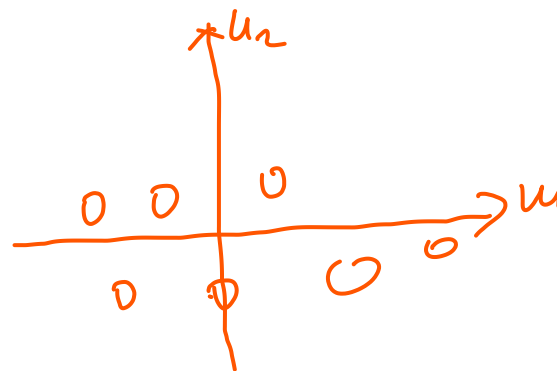
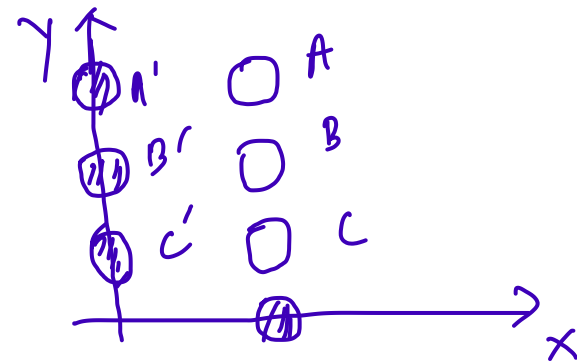


Principal Component Analysis (PCA):

3D $\rightarrow$ 2D



2D $\rightarrow$ 1D



Essence of PCA:

- We project our high-dimensional data to a new coordinate system
 - Choose only a few of these dimensions that explains most of the variations in the data
- axes are orthogonal to each other

PCA:

$$\text{Data } X = \begin{bmatrix} & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \end{bmatrix}_{n \times d} \quad \text{where } n = \# \text{ samples} \\ d = \# \text{ features}$$

↓ PCA

$$X' = Y = \begin{bmatrix} & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \end{bmatrix}_{n \times d'} \quad \text{where } d' \ll d$$

Goal:

- Project the data X on a vector u such that
- the variance of the projected data is maximum ✓
 - u is a unit vector ✓

Projection of data on a vector:

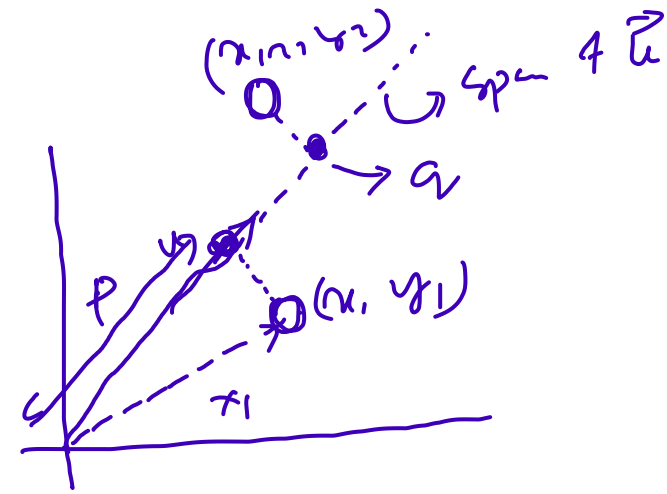
$$X = \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \end{bmatrix}_{2 \times 2} \quad \begin{array}{l} 2 \text{ samples, } u \\ 2 \text{ features} \end{array}$$

$$\|u\| = 1$$

$$p = \text{proj}_u x_1 = \frac{x_1 \cdot u}{\|u\|} = x_1 \cdot u \quad \begin{array}{l} \text{dot product} \\ \text{since } \|u\| = 1 \end{array}$$

$$u \in \mathbb{R}^d, u = \begin{bmatrix} u_1 \\ \vdots \\ u_d \end{bmatrix}_{d \times 1} \quad p = x_1 \cdot u = m x_1 + n y_1 \quad \checkmark$$

$$X = \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \end{bmatrix}_{n \times d} \quad u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_d \end{bmatrix}_{d \times 1}$$



$$x_1 = (x_1, y_1)$$

$$p \text{ proj}_u X = X u = \begin{bmatrix} x_1' \\ x_2' \\ \vdots \\ x_n' \end{bmatrix}_{n \times 1} \quad \begin{array}{l} x_1 u_1 + y_1 u_2 \\ x_2 u_1 + y_2 u_2 \\ \vdots \\ x_n u_1 + y_n u_2 \end{array}$$

P(A): Objective function: $\arg \max_u [\text{Var}(Xu)]$

Constraint $\|u\| = 1$

$\text{Var}(Xu) = u^T \Sigma u$
 $\hookrightarrow$ Covariance matrix $d \times d \rightarrow$ Square & Symmetric
 PSD $\rightarrow$ eigenvalues are nonnegative
 * eigenvectors are orthogonal

eigen value
 $A\vec{v} = \lambda \vec{v}$
 eigen vectors

$\arg \max_u [u^T \Sigma u]$

$\|u\| = 1$

Syn:

$$\boxed{\Sigma u = \lambda u}$$

(u)

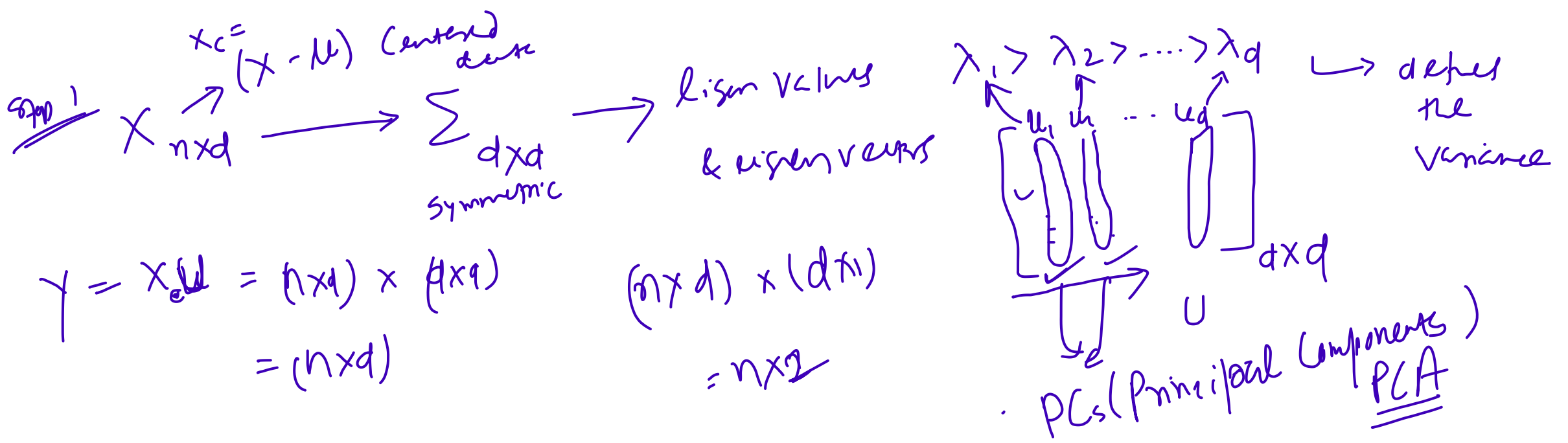
$\hookrightarrow$ EV of $\Sigma_{d \times d}$

* Lagrangian Multiplier ✓
SVM

$$L = f(u) + \lambda g(u)$$

$$\frac{\partial L}{\partial u} = 0$$

$$\frac{\partial L}{\partial \lambda} = 0$$



How to choose d' ?

1st PC (u_1) = $\frac{\lambda_1}{\lambda_1 + \lambda_2 + \dots + \lambda_d}$ λ_1 is r.

two PCs = $\frac{\lambda_1 + \lambda_2}{\lambda_1 + \lambda_2 + \dots + \lambda_d}$

$\frac{\sum_{i=1}^{d'} \lambda_i}{\sum_{j=1}^d \lambda_j} > 0.9$

